Chapter 3

Experiments on Predicting Liver Fat

This chapter records the experiments run to predict the controlled attenuation parameter (CAP; NHANES variable LUXCAPM, dB/m), the FibroScan measure of liver fat that the patient-state model of HepatoTwin estimates from routine data. Fifty-one experiments, run between 5 September and 22 October 2026, varied the learning algorithm, the inputs, the training weights, the treatment of the target, the training data and the evaluated population. Section 3.1 defines how results were measured; Section 3.2 lists every experiment and its result; Section 3.3 describes the model that was selected and its performance; Section 3.4 explains why R^2 stops rising; and Section 3.5 compares the result with published models. Section 3.6 summarises the chapter.

3.1 Evaluation Protocols

The cohort comprised NHANES participants with a complete elastography examination (LUAXSTAT =1) and an observed CAP, aged 12 to 80 years. R^2 is the share of the variation in measured CAP that the predictions explain on participants the model was not trained on. Three protocols were used as the work progressed (Table 3.1). Because each protocol scores a different set of participants, R^2 values from different protocols are not directly comparable, and every result below is labelled with its protocol.

Table 3.1. Evaluation protocols.

Code	Design	Used for
P1	9,021 participants (2017–March 2020) split once 80/20 (seed 42) into 7,216 development and 1,805 test participants; model choice by five-fold cross-validation on the development set (P1-CV), test results as P1-test.	September and early October. The test set was examined repeatedly, so P1-test is an internal comparison.
P2	All 9,021 participants, five-fold cross-validation (seed 42); R^2 on the pooled out-of-fold predictions. Baseline 0.5135.	Model families and new data (October).
P3	15,300 participants from both FibroScan cycles (2017–2020 and 2021–2023). A 20% test set (3,060, stratified by cycle, seed 42) set aside first; model choice by five-fold cross-validation on the remaining 12,240 (P3-CV); test set scored once (P3-test).	Combined model and final selection.

Outcomes are labelled *Adopted* (carried into the model), *Small gain* (below about 0.01, not or not fully adopted), *No gain*, *Worse* and *Diagnostic* (an analysis explaining the model's behaviour rather than a candidate model).

3.2 Experiments

The experiments fall into six phases. Each phase was prompted by the findings of the one before it: once changes to the model stopped helping, the work turned to the target variable, then to new data, and finally to the population in which the model is used.

3.2.1 Phases 1 and 2: baselines, tuning and inputs

The first models used 14 inputs: age, sex, race and ethnicity, BMI, waist circumference, ALT, AST, HbA1c, total and HDL cholesterol, and day-one dietary energy, protein, carbohydrate and fat (Table 3.2).

Table 3.2. Phase 1 and 2 experiments: baselines, tuning and inputs.

ID	Experiment	Result	Outcome
E1	Baseline XGBoost, 14 inputs	P1-test R^2 0.491; RMSE 44.5 dB/m	Adopted
E2	XGBoost against linear regression and random forest	P1-test 0.491; 0.488; 0.486	Small gain
E3	Leakage-free preprocessing and a 40-setting search	Corrected baseline 0.485 → 0.506 (P1-test)	Adopted
E4	14, 19 and 28 inputs (extra blood tests) and 24 model variants	P1-CV 0.489 / 0.496 / 0.501; blend 0.502	Small gain
E5	Reproduction of a supplied model reporting 0.535	On its own split: supplied 0.535, ours 0.541; on our split: 0.511 and 0.514	Diagnostic
E6	36 raw inputs plus 10 engineered features	P1-CV 0.506; P1-test 0.515	Small gain

A linear model nearly matched the tree ensemble (E2), which already suggested that the signal is largely additive and dominated by body size. The first statistically clear gain came from fitting imputation inside each fold and tuning the trees (E3): shallow trees, a low learning rate and row subsampling limit overfitting on a noisy target. Experiment E5 showed that a supplied model's higher R^2 came from an easier test split rather than a better model, and that a single 20% split moves R^2 by about ±0.03.

3.2.2 Phase 3: features, weighting and learners

Phase 3 tested whether reshaping the inputs or the training weights could help (Table 3.3). None changed R^2 by more than 0.003.

Table 3.3. Phase 3 experiments: feature ablations and weighting (P1-CV).

ID	Experiment	Result	Outcome
E7	Engineered-feature ablation	0.5035 / 0.5048 / 0.5032 / 0.5036	Small gain
E8	Single ratios (WHtR, AST/ALT, TG/HDL, TC/HDL)	0.5022–0.5038 against 0.5035	No gain
E9	LDL cholesterol	0.5036 → 0.5020 / 0.5030	No gain
E10	Recursive feature elimination, 46 to 10 inputs	46: 0.504; 25: 0.508; 15: 0.501; 10: 0.485	No gain
E11	Histogram gradient boosting	0.486 against XGBoost 0.505	Worse
E12	Up-weighting extreme CAP values	0.505 → 0.500 / 0.494 / 0.488	Worse

About 20 to 25 inputs carry nearly all the signal, led by waist circumference, waist-to-height ratio and BMI (E10). Derived scores such as the Fatty Liver Index re-express inputs that the trees can already combine (E7–E8).

3.2.3 Phase 4: the target and other model families

Phase 4 first examined the target itself (Table 3.4) and then tried other model families and target treatments (Table 3.5).

Table 3.4. Phase 4 diagnostics of the target.

ID	Experiment	Result	Outcome
E13	Residuals by CAP range (P2)	Slope of prediction on actual 0.51; bias +44 dB/m below 200 and −67 dB/m above 390	Diagnostic
E14	CAP ceiling audit	129 of 9,021 (1.4%) exactly 400 dB/m	Diagnostic
E15	Refit without CAP =400	R^2 0.506 → 0.488; RMSE 44.3 → 43.7 dB/m	No gain
E16	Performance by probe	Within-probe R^2: M 0.419, XL 0.278	Diagnostic

Predictions are pulled towards the mean of about 255 dB/m (E13): the inputs cannot separate very lean or very fatty livers from average ones. The device truncates CAP at 400 dB/m (E14), but too few readings are affected to limit R^2, and removing them lowers R^2 because the outcome variance shrinks (E15).

Table 3.5. Phase 4 experiments: model families and target treatments.

ID	Experiment	Result	Outcome
E17	Probe type (M or XL) as an input	0.5062 → 0.5112	Small gain[a]
E18	XL examinations weighted 1.5× and 2×	0.5112 → 0.5080 / 0.5018	Worse
E19	CatBoost (P2)	0.5093 against XGBoost 0.5135	No gain
E20	Steatosis-class probabilities fed to the regressor	0.5116 / 0.5129	No gain
E21	Linear recalibration of predictions	0.5127	No gain
E22	Quantile (median) regression	0.503 against 0.506; 80% interval covers 74.7%	No gain
E23	Fold-local Lasso and RFE selection	0.499; 0.504	No gain
E24	Extra Trees, random forest, ridge	0.500; 0.498; 0.482	No gain
E25	Neural network (128, 64)	0.453	Worse
E26	Stacking with a ridge meta-learner	0.506 → 0.509	Small gain
E27	Steatosis classification	Binary AUC 0.862; four-class macro F1 0.439	Adopted
E28	HbA1c ablation	Not completed	–

[a]Not used: probe type is recorded during the FibroScan examination itself.

Every model family reached the same ceiling (E19, E24–E26). The binary classification of steatosis (E27) performed well and became the second output of the model; the four-class task failed for the middle grades, which span only 20 and 12 dB/m, less than the model error.

3.2.4 Phase 5: additional data

Phase 4 indicated that the limit lay in the information available, not in the model, so Phase 5 added NHANES data not used before (Table 3.6). Elastography fields and self-reported liver conditions were never used as inputs, because they would leak the target.

Table 3.6. Phase 5 experiments: additional data.

ID	Experiment	Result	Outcome
E29	About 80 more NHANES variables (insulin, CRP, blood count, blood pressure, questionnaires, prescriptions)	0.5135 → 0.5239 (128 inputs, P2)	Small gain
E30	Learning curve (25–100% of training data)	0.488; 0.501; 0.511; 0.512	Diagnostic
E31	2021–2023 FibroScan cycle (6,279 participants)	0.5135 → 0.5170	Small gain
E32	DXA visceral and subcutaneous fat (3,307)	0.5094 → 0.5168	Small gain
E33	Measurement noise within an examination	About 3.7% of CAP variance	Diagnostic
E34	Upper bound with FibroScan-derived inputs	0.518 → 0.524 (probe), 0.529 (stiffness)	Diagnostic
E35	Combined model: both cycles, 81 inputs, reliability weights (P3)	P3-test 0.480 → 0.493; P3-CV 0.520	Adopted
E36	Cohort filters	All 0.518; IQR <40 and ages 12–59: 0.584	Diagnostic
E37	Performance by age group	12–17: 0.43; 18–39: 0.54; 40–59: 0.49; 60+: 0.41	Diagnostic
E38	Improving the 60+ group	0.405 → 0.394–0.408	No gain
E39	Calibrated classifiers, CAP ≥ 248 and ≥ 280	AUC 0.856 and 0.868; calibration error 0.025 → 0.018	Adopted

The largest single gain in the record, +0.010, came from the extra NHANES variables (E29), mainly insulin resistance and diabetes and weight history. The learning curve was flat (E30), so a second cycle of 6,279 participants added only 0.004 (E31). Even visceral fat measured directly by DXA, the strongest known correlate of liver fat, added only 0.007 (E32), and inputs taken from the FibroScan examination itself reached only 0.53 (E34). The combined model of E35 pooled both cycles and down-weighted less reliable examinations; it was adopted.

3.2.5 Phase 6: target population and lean participants

The cohort filters of E36 showed that R^2 is highest among participants aged 12 to 59 with a reliable examination (CAP interquartile range below 40 dB/m). In this group the error itself falls (RMSE 43.6 → 41.6 dB/m) while the spread of CAP does not shrink, so the higher R^2 reflects better predictions, not an easier outcome. On the supervisor's advice this group was adopted as the model's scope, and Phase 6 tried to reach 0.60 within it (Table 3.7).

Table 3.7. Phase 6 experiments within the scope (P3-CV; reference 0.5945).

ID	Experiment	Result	Outcome
E40	Training only on the scope; scope weighted 3×; ages 12–59 only	0.5845; 0.5921; 0.5883	Worse
E41	Habitual diet from the 24-hour recall	0.5930	No gain
E42	Medical history and prescription count	0.5949	No gain
E43	Fasting subsample only	0.597 against 0.592 (test: 0.550 against 0.560)	No gain
E44	Median imputation with indicators; MICE	0.5961; 0.5938	No gain
E45	Pseudo-Huber loss; absolute-error loss	0.5940; 0.5893	No gain
E46	Five further tree settings	0.5915–0.5960	Adopted[b]
E47	Ensembles of 2–5 variants	Best 0.5969	Small gain
E48	Lean-only classifier (BMI <28)	Lean AUC 0.804 against 0.809	Worse
E49–E51	Lean weighting, non-obese training, blend	Lean AUC 0.807–0.808	No gain

[b]The deepest setting had the highest test R^2 of all candidates (Section 3.3).

All variants scored between 0.584 and 0.597, a spread smaller than the variation between cross-validation folds (0.546 to 0.617). Training on the scope alone was worse than training on everyone (E40): participants outside the scope still carry information about how the inputs relate to CAP. Discrimination in lean participants (E48–E51) fell in every BMI band, and the Fatty Liver Index fell by a similar amount, because within a narrow BMI band the strongest predictor, body size, hardly varies.

3.3 Final Model and Results

The selected model is an XGBoost regressor [1] with 3,000 trees, depth 5, learning rate 0.005, minimum child weight 15 and column subsampling 0.4, trained on the 12,240 training participants of both FibroScan cycles. It uses 81 routine inputs (body measurements, standard blood tests, blood pressure and a short questionnaire) and six derived scores: waist-to-hip ratio, HOMA-IR, the triglyceride-glucose index, mean systolic and diastolic blood pressure, and one-year weight change. Examinations with a CAP interquartile range of 40 dB/m or more are down-weighted (weight 40/IQR, at least 0.25) rather than dropped. Two isotonic-calibrated classifiers estimate the probability that CAP is at least 248 and at least 280 dB/m.

The tree settings were chosen by scoring each of the 15 Phase 6 candidates on the in-scope members of the held-out test set; the deepest setting scored highest (0.5622 against 0.5582 for the reference). It also has the highest full-population cross-validation R^2 of the 12 settings compared, and repeated cross-validation, which played no part in the choice, confirms the gain (0.593 against 0.590). Because the test set informed the choice, the test figures carry slight selection optimism.

Table 3.8. Selected model on the held-out test set (P3-test). The reference uses the 36 core inputs on the same participants.

Test group	n	R^2	Reference R^2	RMSE (dB/m)
Scope: ages 12–59, reliable reference	1,236	0.562	0.553	41.4
Ages 12–17	209	0.498	0.482	36.2
Ages 18–39	512	0.518	0.514	42.7
Ages 40–59	515	0.522	0.506	42.1
Outside the scope: ages 12–59, all examinations	2,040	0.515	0.511	–
Outside the scope: all participants	3,060	0.496	0.487	43.5

Within the scope the held-out test R^2 is 0.562 (95% bootstrap interval 0.517–0.601; Table 3.8), and 0.593 in five-fold cross-validation over all 15,300 participants repeated with three seeds. The value is lower within each age band than in the whole scope because CAP varies less within a band. Further checks did not change the picture (Table 3.9): the result holds when weighted to the US population, and a model trained on 2017–2020 alone predicted the 2,415 in-scope participants of 2021–2023 with R^2 0.578.

Table 3.9. Selected model within the scope, with 95% bootstrap intervals.

Measure	Held-out test	Survey-weighted	Temporal (2021–2023)
CAP R^2	0.562 (0.517–0.601)	0.554 (0.493–0.609)	0.578 (0.549–0.603)
CAP RMSE (dB/m)	41.4 (39.3–43.6)	42.6 (39.5–45.7)	42.9 (41.4–44.5)
AUC, CAP ≥ 248	0.881 (0.863–0.899)	0.877 (0.851–0.901)	0.884 (0.870–0.896)
AUC, CAP ≥ 280	0.879 (0.860–0.898)	0.878 (0.849–0.905)	0.890 (0.878–0.904)

On the 1,100 in-scope test participants with every input to both scores, the CAP ≥ 248 classifier reached AUC 0.881 against 0.861 for the Fatty Liver Index [2] and 0.836 for the Hepatic Steatosis Index [3]; the paired differences (+0.020 and +0.046) both exclude zero.

3.4 Why R^2 Plateaus

Nine independent lines of evidence show that the remaining variation in CAP cannot be explained by routine clinical measurements:

1. Every algorithm tested (linear regression, random forest, Extra Trees, histogram boosting, XGBoost, CatBoost, a neural network and stacking) reaches R^2 between 0.45 and 0.52 on the same inputs (E2, E11, E19, E24–E26).
2. Feature engineering, selection and ablation change R^2 by at most 0.003 (E7–E10, E23).
3. The learning curve is flat: 6,279 additional participants add 0.004 (E30, E31; Figure 3.1).
4. Direct DXA measurement of visceral fat adds 0.007 (E32).
5. About 80 additional NHANES variables add 0.010 (E29).
6. Even inputs taken from the FibroScan examination itself reach only 0.53 (E34).

7. Noise within one examination is about 4% of the CAP variance (E33), so the unexplained variance reflects real differences between people with similar measured profiles.
8. Within the scope, every change to inputs, imputation, loss, tuning, training population and ensembling scored between 0.584 and 0.597, a spread smaller than the variation between folds (E40–E47).
9. CAP is itself an imperfect measure of liver fat: two operators agree with a concordance of 0.82 [4], and CAP correlates with MRI-PDFF at r = 0.66 [5], sharing about 44% of its variance with the reference measure.

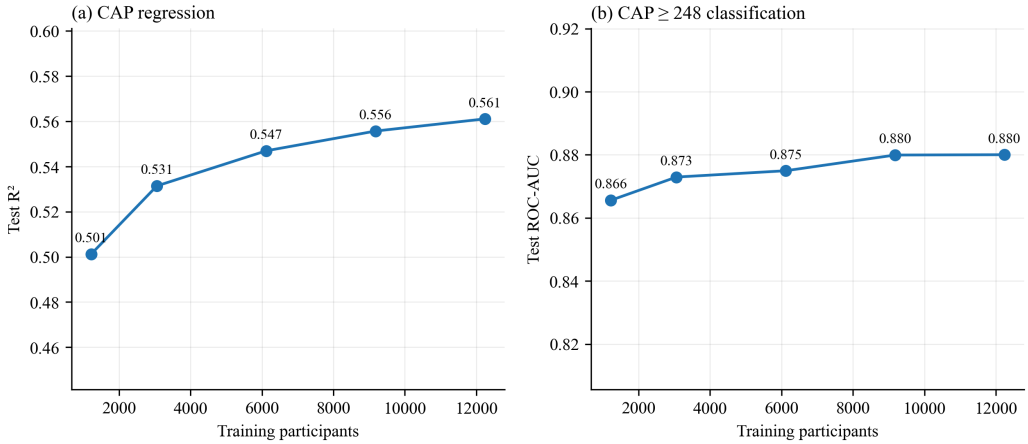


Figure 3.1. Learning curve of the selected settings: held-out test R^2 and AUC within the scope (n = 1,236) against the number of training participants. R^2 rises from 0.501 with 1,224 participants to 0.556 with 9,180 and only to 0.561 with all 12,240.

Two causes dominate. First, about half of liver fat is genetic: in a prospective twin study the heritability of steatosis measured by MRI-PDFF was 0.52 (95% CI 0.31–0.73) [6]. Variants such as PNPLA3 I148M change liver fat substantially at the same body size, and NHANES 2017–2023 contains no genotypes; lifelong diet, an accurate alcohol history and muscle mass are measured poorly or not at all. Two people with the same weight, waist and blood tests can therefore have very different livers. Second, the model is scored against a single CAP reading. With a reproducibility of about 0.82, even a model that knew each patient's true liver state could not exceed an R^2 of the order of 0.8. The heritable share and the other unmeasured factors lower the attainable value further, to around 0.6 for routine data. The in-scope result of 0.562 (0.593 in cross-validation) is close to that ceiling: the plateau is a property of the data, not of the model.

This is also why HepatoTwin relies on the calibrated probabilities of steatosis rather than on the exact CAP value. They discriminate well (AUC 0.88) and are well calibrated, and they outperform the established clinical scores on the same participants.

3.5 Comparison with Published Models

Published models fall into two groups: those predicting a continuous liver-fat value, which report R^2, and NHANES models classifying steatosis, which report AUC. No NHANES study reporting R^2 for CAP was found, so this work provides the first such benchmark.

Table 3.10. Continuous liver-fat prediction from routine data.

Study	Population	Reference	Inputs	R^2
UK Biobank [7]	40,533 UK adults	MRI-PDFF	37 body and blood measures	0.37
UK Biobank [7]	9,723 UK adults	MRI-PDFF	Plasma metabolomics	0.34
UK Biobank [7]	46,000 UK adults	MRI-PDFF	DXA images (not routine)	0.63–0.64
Asaturyan et al. [8]	28,796 UK adults	MRI-PDFF	Fatty Liver Index alone	0.28–0.36
Asaturyan et al. [8]	As above	MRI-PDFF	12 variables incl. diagnoses	0.54
Petäjä et al. [9]	378 Finnish adults	^{1}H-MRS	Includes a specialist marker	0.44 / 0.49[c]
Liu H. et al. [10]	272, obesity clinic	CAP	BMI, neck, TG, C-peptide	0.63–0.65[d]
Haas et al. [11]	UK Biobank	MRI	BMI and 24 clinical factors	≈0.34[e]
Fatty Liver Index, this cohort	13,502 / 5,548	CAP	FLI (linear fit)	0.44 / 0.50
This study	15,300 NHANES	CAP	81 routine inputs	0.56 test; 0.59 CV

\multicolumn{5}{p{14cm}}{ [c]Adjusted, discovery / validation. [d]Validation sample of a small, selected cohort with high CAP (median 320–335 dB/m against 251 in NHANES), which widens the outcome range and raises R^2. [e]Reported as r = 0.58. Kotronen et al. [12] and Chen et al. [13] report no R^2. This study: within the scope; 0.50 for all participants.}

Table 3.11. NHANES steatosis classification models.

Study	Outcome	Reported	This study
Du et al. [14]	MASLD, CAP ≥ 280	AUC 0.815–0.842	0.879 (0.868)
Liu L. et al. [15]	CAP ≥ 302	AUC 0.859; FLI 0.838	0.888 (0.872)
Zhang et al. [16]	CAP ≥ 302	AUC 0.874	0.888 (0.872)
Boullion et al. [17]	CAP ≥ 238	Accuracy 79.6%	0.881 (0.857), CAP ≥ 248
Zhu et al. [18]	Asian, BMI <28	Macro AUC 0.874 (three grades)	0.799 test; 0.823 CV
\multicolumn{4}{p{14cm}}{ This study: AUC within the scope, with all 3,060 test participants in brackets. The published studies cover adults of all ages, so the bracketed values are the closer comparison.}			

Within its scope, the model's held-out R^2 of 0.56 (0.59 in cross-validation) is comparable to or above published routine-data models of liver fat (0.34–0.54; Table 3.10), including 0.37 in UK Biobank and 0.49 in validation for a spectroscopy equation that needs a specialist blood marker. Across all ages its R^2 of 0.50 still exceeds the UK Biobank routine-data model. Higher values come only from a small selected sample (0.65) or from imaging inputs (0.63–0.64). The studies differ in reference measure and population, so these values are indicative rather than directly comparable; the claim that can be made is “comparable to or above published routine-data models”, not “state of the art”. In classification the model matches or exceeds every NHANES model (Table 3.11). The exception is lean Asian participants, where its AUC (0.80) is below the three-grade macro AUC of Zhu et al. (0.874), a metric that is not directly comparable and that rests on a training set of about 150,000 Chinese adults.

3.6 Summary

Fifty-one experiments raised R^2 from 0.491 for the first 14-input model to 0.562 on the held-out test set within the model's scope (ages 12 to 59, reliable reference examination), 0.593 in repeated cross-validation, and 0.578 on participants examined two to six years after the training data. Changes of algorithm, features, loss or tuning each moved R^2 by about 0.01 or less; the gains came from adding information (more NHANES variables and a second cycle) and from defining the population in which the model is used. The plateau near 0.56–0.59 is set by the data: about half of liver fat is heritable and unmeasured, and CAP is itself a noisy measure. At this level the model is comparable to or above published routine-data models, and its calibrated steatosis probabilities, which HepatoTwin uses, outperform the Fatty Liver Index and the Hepatic Steatosis Index.

The main limitations are that all validation is internal to NHANES; that the final settings were chosen with the help of the test set, which makes the test R^2 slightly optimistic; that patients aged

60 and over are outside the scope; and that the data are cross-sectional, so the model describes the current state and cannot by itself predict the response to a change in diet.

References

1. Chen T, Guestrin C. XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* 2016:785–794.
2. Bedogni G, et al. The Fatty Liver Index: a simple and accurate predictor of hepatic steatosis in the general population. *BMC Gastroenterology* 2006;6:33.
3. Lee J, et al. Hepatic steatosis index: a simple screening tool reflecting nonalcoholic fatty liver disease. *Digestive and Liver Disease* 2010;42(7):503–508.
4. Ferraioli G. Quantitative assessment of liver steatosis using ultrasound controlled attenuation parameter (Echosens). *Journal of Medical Ultrasonics* 2021;48:489–495. doi:10.1007/s10396-021-01106-1
5. Ahn SB, Jun DW, Kang B, et al. Optimal cutoff value for assessing changes in intrahepatic fat amount by using the controlled attenuation parameter in a longitudinal setting. *Medicine* 2018;97(50):e13636. doi:10.1097/MD.0000000000013636
6. Loomba R, Schork N, Chen C, et al. Heritability of hepatic fibrosis and steatosis based on a prospective twin study. *Gastroenterology* 2015;149(7):1784–1793. doi:10.1053/j.gastro.2015.08.011
7. Machine learning across multiple imaging and biomarker modalities in the UK Biobank improves genetic discovery for liver fat accumulation. *medRxiv preprint* 2024. doi:10.1101/2024.01.06.24300923
8. Asaturyan HA, Basty N, Thanaj M, Whitcher B, Thomas EL, Bell JD. Improving the accuracy of fatty liver index to reflect liver fat content with predictive regression modelling. *PLoS ONE* 2022;17(9):e0273171. doi:10.1371/journal.pone.0273171
9. Petäjä EM, Zhou Y, Havana M, Hakkarainen A, Lundbom N, Ihalainen J, Yki-Järvinen H. Phosphorylated IGFBP-1 as a non-invasive predictor of liver fat in NAFLD. *Scientific Reports* 2016;6:24740. doi:10.1038/srep24740
10. Liu H, Li X, Han X, et al. Simple surrogate equations to predict controlled attenuation parameter values for screening non-alcoholic fatty liver disease in a Chinese population. *Frontiers in Medicine* 2022;9:894895. doi:10.3389/fmed.2022.894895
11. Haas ME, Pirruccello JP, Friedman SN, et al. Machine learning enables new insights into genetic contributions to liver fat accumulation. *Cell Genomics* 2021;1(3):100066. doi:10.1016/j.xgen.2021.100066
12. Kotronen A, Peltonen M, Hakkarainen A, Sevastianova K, Bergholm R, et al. Prediction of non-alcoholic fatty liver disease and liver fat using metabolic and genetic factors. *Gastroenterology* 2009;137(3):865–872. doi:10.1053/j.gastro.2009.06.005
13. Chen J, Yue J, Fu J, et al. A prediction model of liver fat fraction and presence of non-alcoholic fatty liver disease (NAFLD) among patients with overweight or obesity. *Endocrine Journal* 2023;70:977–985. doi:10.1507/endocrj.EJ23-0227
14. Du S, Yu H, Du J, Xu Y. Machine learning models to predict metabolic dysfunction-associated steatotic liver disease (MASLD) with simple anthropometric and biochemical variables: a cross-sectional study in US population. *International Journal of Hepatology* 2026;2026:8221645. doi:10.1155/ijh/8221645
15. Liu L, Lin J, Liu L, Gao J, Xu G, Yin M, Liu X, Wu A, Zhu J. Automated machine learning models for nonalcoholic fatty liver disease assessed by controlled attenuation parameter from the NHANES 2017–2020. *Digital Health* 2024;10:20552076241272535.
16. Zhang Y, Liu X, Zhang X, Fei Y, Li X. Machine learning-based prediction of metabolic dysfunction-associated steatotic liver disease using National Health and Nutrition Examination Survey (NHANES) data. *PLoS ONE* 2025;20(11):e0335656. doi:10.1371/journal.pone.0335656

17. Boullion J, Husein A, Agrawal A, et al. Machine learning–based biomarker identification for early diagnosis of metabolic dysfunction–associated steatotic liver disease. *The Journal of Clinical Endocrinology \& Metabolism* 2025;110(11):e3866–e3877. doi:10.1210/clinem/dgaf111

18. Zhu Y, Wang Y, Zhang S, Yang J, Zhang F, Liu Y, et al. Hepatic steatosis severity prediction in nonobese individuals: machine learning model development and validation. *Journal of Medical Internet Research* 2026;28:e82529.